B.Tech (1st Semester) **GROUP-C** **Batch (2026-30)**

Course Code-100103P **Engineering Chemistry Lab** **0 0 2 1**

Mapped CO(s) Number	Practical/ Lab Session Outcomes (LSOs)	Laboratory Experiment/ Practical Titles
CO5	*LSO 1.1* Calibrate and operate a digital pH meter accurately *LSO 1.2* Perform acid-base titration to determine the molarity and pH of the unknown acid.	LE- 1 Determination of pH of a given acid solution using a standard sodium hydroxide solution.
CO5	*LSO 2.1* Prepare a standard EDTA solution *LSO 2.2* Calculate Total, Temporary, and Permanent hardness in terms of $CaCO_3$ equivalents.	LE- 2 Estimation of hardness of water by the EDTA method.
CO5	*LSO 3.1* Operate the ion exchange column. *LSO 3.2* Evaluate the efficiency of the softening process by comparing hardness before and after treatment.	LE- 3 Removal of hardness of water by ion exchange method.
CO5	*LSO 4.1* Perform the titration to estimate hardness *LSO 4.2* Determine the Hardness of water.	LE- 4 Estimation of the hardness of water by the alkalinity method
CO4	*LSO 5.1* Set up a potentiometric cell using reference and indicator electrodes. *LSO 5.2* Determine the equivalence point of a redox titration (e.g., Fe^{2+} vs $K_2Cr_2O_7$) from an EMF vs Volume graph.	LE- 5 Potentiometric determination of redox potentials and emf.
CO3	*LSO 6.1* Detect specific functional groups (such as Alcohols, Phenols, Aldehydes, or Acids) using characteristic chemical reagents. *LSO 6.2* Use the physical properties to narrow down the identity of an unknown organic sample.	LE- 6 Identification of organic substances and their functional groups.
CO5	*LSO 7.1* Perform Standard tests *LSO 7.2* Interpret the results to identify the sources of adulteration	LE- 7 Test of adulteration of fat butter, sugar, turmeric powder and chilli powder
CO4	*LSO 8.1* Use standard analytical procedures. *LSO 8.2* Determine the saponification value and acid value of the given oil sample *LSO 8.3* Observe and record the colour change of indicators to find the exact endpoint of the reaction.	LE- 8 Saponification/acid value of an oil.
CO5	*LSO 9.1* Estimate the residual chlorine content in a water sample using colourimetric analysis. *LSO 9.2* Demonstrate safe laboratory practices while handling water samples and chemical reagents.	LE- 9 Estimation of Residual Chlorine in a water sample by Colourimetric analysis.
CO2	*LSO 10.1* Prepare a sample for the FT-IR machine (such as using a liquid film or a solid pellet).	LE- 10 Identification of organic functional groups in an unknown sample using FT-IR Spectroscopy.

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	LSO 10.2 Identify key peaks and valleys in an IR spectrum to recognize groups like –OH (Alcohols), C=O (Carbonyls), and -NH2 (Amines).	
CO2	*LSO 11.1* *Perform* TLC to separate amino acids in a mixture. *LSO 11.2* *Determine* R_f values and *identify* amino acids	LE- 11 Qualitative analysis of a mixture of amino acids by TLC and visualisation under UV light.
CO4	*LSO 12.1 Operate the Flash Point apparatus* *LSO 12.2 Determine the flash point and pour.*	LE- 12 Determination of Flash point, Pour point.
CO3	*LSO 13.1* Use standard polymerisation techniques. *LSO 13.2* Prepare the resin. *LSO 13.3* Follow safe laboratory practices during polymer synthesis and handling of chemicals	LE- 13 Synthesis of polymer like Urea-formaldehyde resin.